

Modernizing the Flipped Classroom: Replacing Lecture Time With Asynchronous Spaced Retrieval



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Abstract

Background: Traditional flipped classrooms require learners to watch video lectures before attending class. This can pose time burdens on both the instructor and the learner.

Objective: We examined whether delivering traditional lecture content via asynchronous spaced retrieval practice could teach learners a new topic without a lecture.

Method: Twenty-eight participants watched a lecture and then used an app for 4 weeks to answer questions about the lecture and a new topic not accompanied by a lecture. A final test featuring new questions assessed knowledge transfer.

Results: During app usage, participants started with lower accuracy on nonlectured than lectured questions, but after just a week, this difference disappeared. Overall, learning took less time than watching a lecture. On the final test of knowledge transfer, participants performed equally well on lectured and nonlectured questions.

Conclusion: Spaced retrieval practice alone can teach learners new concepts in less time than a traditional lecture.

Teaching Implications: In a flipped classroom, delivering foundational course content through spaced retrieval sessions may offer a new method of teaching that fosters student engagement through interactive learning while reducing the time needed for instructors to create and maintain video lectures.

Keywords

flipped classroom design, retrieval practice, lecture time, instructional technology, knowledge transfer

Flipped classrooms reverse traditional learning environments by requiring students to engage in significant learning activities before attending class, usually through readings or videos lectures, allowing more in-class time for interactive activities. Flipped classrooms are becoming more common in higher education. Still, they can come at a time burden to both instructors who must create and maintain content and learners who must find independent time to engage meaningfully with the content (c.f. O'Flaherty & Phillips, 2015). *Spaced retrieval* is an evidence-supported learning strategy that shows great promise to maximize time and learning in the flipped classroom. This involves studying information by repeatedly testing oneself on it (i.e., *retrieving*) following some period of elapsed time since the last exposure (i.e., *spacing*). Spaced retrieval improves long-term memory retention compared to other study methods such as re-reading, note-taking, or non-spaced (massed) retrieval practice (Higham et al., 2022; McHugh et al., 2021). Additionally, when time spent engaging in spaced retrieval is optimized by a machine learning algorithm, spaced retrieval can greatly reduce the amount of time learners need to establish strong long-term memories relative to other methods, such as traditional flashcard studying (McHugh et al., 2021). Thus, implementing an optimized practice of spaced retrieval in a flipped classroom can potentially reduce the time spent learning while simultaneously boosting long-term retention.

The spaced retrieval practice strategy is not new to higher education. For example, frequent low-stakes quizzes and

cumulative exams are commonly included in college courses. Further, students tend to seek out retrieval practice tools. For instance, a recent survey of undergraduates found that about 78% used digital flashcards, primarily Quizlet and Anki, to study material learned in their college classes (Zung et al., 2022). In general, prior research has focused on how students use flashcards to support their understanding of classroom lectures (e.g., Wissman et al., 2012; Zung et al., 2022) rather than whether they are appropriate for delivering lecture content in a flipped classroom setting.

In addition to time efficiency, another critical point to consider is whether implementing a schedule of asynchronous spaced retrieval practice promotes learning beyond simple memorization and builds transferable knowledge that the learner can use to answer new questions about the learned

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topics. A recent meta-analysis found that, after learning material through retrieval practice, knowledge transfer is most likely when studied questions are nearly identical to tested questions, the learner engages in elaborative retrieval practice when studying, or when the learner scores highly on their first retrieval-practice attempt (Pan & Rickard, 2018). Though transfer effects on a final test can be elusive in real-world contexts because these criteria are not always met, research shows that they are obtainable with increased numbers of retrieval attempts during the learning phase (Vaughn & Rawson, 2011, 2014). Thus, an important consideration in the present research is whether the knowledge acquired via spaced retrieval practice will promote successful knowledge transfer.

In the present small-scale study, we examined a novel application of spaced retrieval practice. Specifically, we tested whether spaced retrieval could teach students foundational course content in the absence of a lecture. Further, we explored whether participants, after learning via retrieval practice, would demonstrate knowledge transfer on a final memory test. If successful, course content could be delivered to students via retrieval practice asynchronously and without an accompanying lecture, simultaneously reducing time burdens on both the student and instructor.

For this effort to succeed, it is important to ensure that the time students spend learning asynchronously via retrieval practice does not exceed the time students would have otherwise spent attending the relevant lecture. To replace lecture time with asynchronous retrieval practice in a time-efficient way, we employed a mobile application called Blank Slate that delivers an individualized program of spaced retrieval. Through its question-answer format, the app prompts learners to retrieve information only when they are at risk of forgetting it, as determined by a machine learning algorithm. Blank Slate has been shown in prior research to help maintain ceiling levels of memory performance for about one-third of the time students take to engage in traditional flashcard studying (McHugh et al., 2021). The app's individualized algorithm distinguishes it from other digital flashcard tools like Anki and Quizlet, but similar results could likely be obtained with those free platforms. Additional information about the Blank Slate app is provided in OSF materials (Gordon et al., 2025).

In this study, 28 participants first watched one of two 27min recorded lectures on unfamiliar topics in psychology. For the subsequent 4 weeks, they used the app to answer questions about the viewed and nonviewed lectures, receiving feedback after each incorrect answer. After 4 weeks, all participants completed a memory test presenting related but new multiple-choice questions and four short essay questions, all evenly divided between the two topics. We examined the final test performance for the two topics to compare outcomes for lectured material and material acquired solely through retrieval practice.

Method

Design

We conducted a within-subjects study where participants were exposed to two conditions: app-based retrieval practice

targeting lectured and nonlectured content. Performance was measured at five time points. This research was approved by the IRB at the first author's institution.

Participants

We used G*Power (Faul et al., 2007) to determine the minimum sample size needed to detect a medium effect ($f = 0.15$) for each of our two primary analyses. To achieve at least 80% power for our 2×5 repeated measures analysis of variance (ANOVA) on learning curves, with an alpha of .05 and assuming a 0.80 correlation among repeated measures, the analysis suggested a minimum sample size of 24 participants. To achieve at least 80% power to detect effects for our paired-samples *t*-tests on final test scores, the analysis suggested a minimum sample of 27 participants. This sample may seem small but is due to our use of a within-subjects design. In comparison, the recommended sample size for a between-subjects design was 104 participants.

In anticipation of dropout, we recruited 34 young adults to participate in the study through flyers posted on the campus of the first author and in the surrounding community. Screenings ensured participants had no formal instruction in general psychology content. Participants who completed the study received \$70, and we prorated compensation for those who ended participation early. Six participants did not complete the study, resulting in 28 whose data were included in all final analyses ($M_{\text{age}} = 20.89$, $SD_{\text{age}} = 4.43$, 16 female). One participant identified as American Indian or an Alaskan Native, four as Asian, three as Black or African American, 16 as White, and four identified as more than one race. Two participants identified as Hispanic or Latino. All participants reported normal or corrected to normal vision. Three participants disclosed a diagnosed learning disability. Four participants reported that English was a second language. We examined data from participants in these latter two categories as potential outliers before inclusion in the analyses.

Materials and Procedure

The study consisted of one face-to-face session, 4 weeks of app use, and a final test. Participants attended an individual 1-hr orientation session scheduled in person or via Zoom. After providing informed consent and demographic information, participants were randomly assigned to watch one of two 27min lecture videos, each featuring PowerPoint slides with voice narration. The lectures were developed from Lumen Learning's *Introduction to Psychology* instructor resources (Spielman et al., 2020), and each presented condensed content that the first author typically covers in a 50min classroom lecture. One lecture presented material on social/personality psychology, and the other presented material on psychological disorders. App accuracy on the first attempt at each question set in the non-lectured condition indicated that the social psychology content was easier for participants, though the difference

was not statistically significant ($M_{\text{social}} = 0.35$, $SD = 0.16$; $M_{\text{disorders}} = 0.26$, $SD = .09$), $t(26) = 1.85$, $p = .076$, $d = 0.70$.

Participants took paper-and-pencil notes during the video lecture as they would during a class. Following the video, participants discarded their notes because they were not needed for the study. Finally, they downloaded the mobile app and completed their first retrieval practice session.

Over the following 4 weeks, participants were prompted via push notifications and email every 1 to 3 days, per the algorithm, to open the app. To keep the time burden light and facilitate adherence, during each app session, they answered up to 15 randomized questions about content from the lectured and nonlectured topics and received corrective feedback after each incorrect attempt. Aside from correcting inaccurate responses, no additional information regarding the participants' current or past performance was provided. Participants attempted each question until they answered it correctly. On average, participants answered lectured questions 8.76 times per session ($SD = 3.58$) and nonlectured questions 9.46 times per session ($SD = 3.36$). The full question bank consisted of a total of 50 multiple-choice questions: 25 questions that comprehensively covered the content from each video lecture. These materials were not piloted, but early participants were monitored for potential ceiling effects. An example from the social/personality content:

"Which of Freud's defense mechanisms is characterized by an individual switching unacceptable impulses into their opposites, such as displaying exaggerated friendliness when they are feeling angry?"

- A. Regression
- B. Reaction formation
- C. Projection
- D. Displacement

Researchers monitored app usage and emailed personal reminders to any participants who had not logged in for 3 days and dropped participants ($n = 6$) who did not respond, which resulted in our final sample of 28 participants.

The day after completing 4 weeks of app use, participants were emailed a link to a two-part memory test administered via Qualtrics. In part one, participants answered 50 questions (25 per content type) directly targeting the details presented via Blank Slate. These questions assessed the application of learning rather than rote memorization of the content. For example, the previous question about Freud was changed to the following:

"Which of the following examples demonstrates Freud's reaction formation defense mechanism?"

- A. An adult displays extreme friendliness when they are feeling angry

B. A 10-year-old boy sucks his thumb after a stressful social encounter at school

C. A man accuses his spouse of being too anxious when he himself is feeling anxious

D. A partner denies evidence of a loved one's affair

After finishing the multiple-choice test, participants answered four open-ended questions in 250 words or less: two on the topic of social/personality (e.g., Describe what you know about the results of the Stanford Prison Experiment.) and two on the topic of disorders (e.g., What causes some people to develop schizophrenia? Please answer as completely as possible.). Following the completion of the final test, participants were contacted for debriefing and compensation.

Data Scoring: The Learning Phase

Estimated Learning Curves. Because the app's spaced-retrieval algorithm prevents participants from reviewing all items in each learning session, we could not compute traditional learning curves. Instead, we estimated learning curves by determining cumulative accuracy proportions for each participant and for each of the two topics (lectured vs. nonlectured content) across the span of 28 days. That is, for each day of the study, we calculated each participant's accuracy on all their most recent responses to each of the 25 questions on lectured content and each of the 25 questions on nonlectured content. For example, a participant's accuracy scores for lectured content on day 14 of the study would be calculated by searching for their most recent viewing of each of the 25 lectured questions and calculating their proportion correct for those 25 questions (when not all 25 had been answered yet, the denominator adjusted accordingly).

App Usage. Questions administered through the mobile app were automatically scored as correct or incorrect; no partial credit was given. To examine within-subjects differences in app usage for questions that were lectured versus nonlectured, we computed for each participant: (1) the total number of questions completed for lectured and nonlectured items, (2) the average seconds spent answering each question for lectured and nonlectured content, and (3) the average minutes spent per week answering lectured and nonlectured content. Concerning the first variable, the total number of questions answered, note that although each participant answered 15 questions per session, they were required to attempt each question repeatedly until they answered it correctly. Therefore, the total number of questions answered could fluctuate based on each participant's accuracy.

Data Scoring: The Final Memory Test

Multiple Choice. Multiple-choice questions on the final test were scored as correct or incorrect; no partial credit was given. Two scores for the final multiple-choice test were calculated for each participant: a proportion of correct answers for the 25 lectured questions and for the 25 nonlectured questions.

Short Essay. Short essay questions were scored on a full point scale from 0 to 3. A score of 0 was assigned when no answer was provided or when completely incorrect answers were provided. A score of 1 was assigned when one correct detail was provided without additional explanation/context. A score of 2 was assigned when two or more correct details were provided with no explanation/context or one detail was provided with the correct explanation/context. A score of 3 was assigned when two or more details were provided with additional explanation/context. The “explanation/context” referred to in this scoring system includes definitions of provided terms, descriptive narration of events, and/or providing examples of terms or events. The first and third authors independently scored all 112 written responses (28 participants $\times$ 4 essays each), blind to which essays came from the lectured and nonlecture conditions and had an agreement rate of 98.21%. The two disagreements were discussed and resolved. Two short essay scores were calculated for each participant: an average accuracy score for the two questions on lectured content and an average score for the two questions on nonlectured content.

Results

The raw data collected during this study are available via OSF (Gordon et al., 2025). In the following analyses, an alpha level of .05 was used to determine statistical significance. Effect sizes are reported using Cohen’s d and η^2 . For Cohen’s d , values of 0.2, 0.5, and 0.8 are interpreted as small, medium, and large effect sizes, respectively (Cohen, 1988). For η^2 , values of 0.01, 0.06, and 0.14 are considered small, medium, and large, respectively (Cohen, 1973).

The Learning Phase

To assess app accuracy over time, we conducted a 2 (content type: lectured vs. nonlectured) $\times$ 5 (time point: day 1, day 8,

day 15, day 22, day 28) repeated-measures ANOVA and found a significant main effect of time point, $F(4, 108) = 33.98$, $p < .001$, $\eta^2 = 0.37$, and a significant time point by content type interaction, $F(4, 108) = 7.20$, $p < .001$, $\eta^2 = 0.04$. The main effect of instruction type was not significant, $F(1, 27) = 2.24$, $p = .146$, $\eta^2 = 0.01$. Post hoc tests using the Holm-Bonferroni correction were used to probe the interaction. On day 1 of the study, participants had higher accuracy on app questions related to lectured content than nonlectured content, $t(27) = 4.53$, $p < .001$, $d = 0.75$ (see Figure 1). By day 8, participants’ accuracy on lectured versus nonlectured questions no longer differed, $t(27) = 1.91$, $p = .878$, $d = 0.31$, and there continued to be no difference in accuracy for lectured versus nonlectured questions on day 15 [$t(27) = 0.74$, $p > .999$, $d = 0.12$], day 22 [$t(27) = 0.27$, $p > .999$, $d = 0.04$], and day 28 [$t(27) = 0.47$, $p > .999$, $d = 0.08$].

We conducted paired-samples t -tests to examine differences in app usage between lectured versus nonlectured content for the average minutes spent per week answering questions, the average seconds spent per question, and the total number of questions completed (see Table 1). Participants spent more minutes per week answering app questions on nonlectured content compared to lectured content, $t(27) = 2.66$, $p = .013$, $d = 0.50$. Participants also spent more seconds per question answering questions on nonlectured compared to lectured content, $t(27) = 2.08$, $p = .047$, $d = 0.39$. There was no significant difference in the total number of questions that participants completed for lectured versus nonlectured content, though results trended toward more questions being reviewed for nonlectured content, $t(27) = 1.80$, $p = .084$, $d = 0.34$.

The Final Memory Test

A paired-samples t -test on final multiple-choice items showed no significant difference in participants’ performance for questions on lectured ($M = 0.70$, $SD = 0.25$) versus nonlectured content ($M = 0.69$, $SD = 0.24$), $t(27) = 0.47$, $p = .319$, $d = 0.09$ (see Figure 2). Similarly, a paired-samples t -test on the final short essay test did not show a significant difference in participants’ average accuracy for questions on lectured ($M = 1.79$, $SD = 0.83$) versus nonlectured content ($M = 1.61$, $SD = 0.61$), $t(27) = 1.20$, $p = .119$, $d = 0.23$.

Table 1. Means (SD) for App Usage Metrics for Lectured and Nonlectured Content.

	Lectured	Nonlectured
Total number of questions completed on the app (includes repeats after questions were answered incorrectly)	156.32 (49.09)	172.00 (48.06)
Minutes spent per week completing questions ^a	7.92 (3.28)	9.05 (3.94)
Seconds spent per question ^a	15.05 (6.87)	15.89 (6.68)

^aSignificant comparison, $p < .05$.

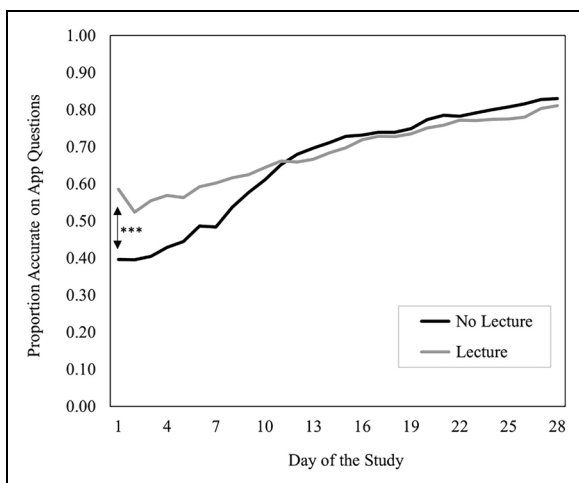


Figure 1. Estimated learning curves for the 4-week period of app use. Note. Accuracy was significantly different between the lecture and no lecture conditions on day 1, but not for any subsequent comparisons. *** $p < .001$.

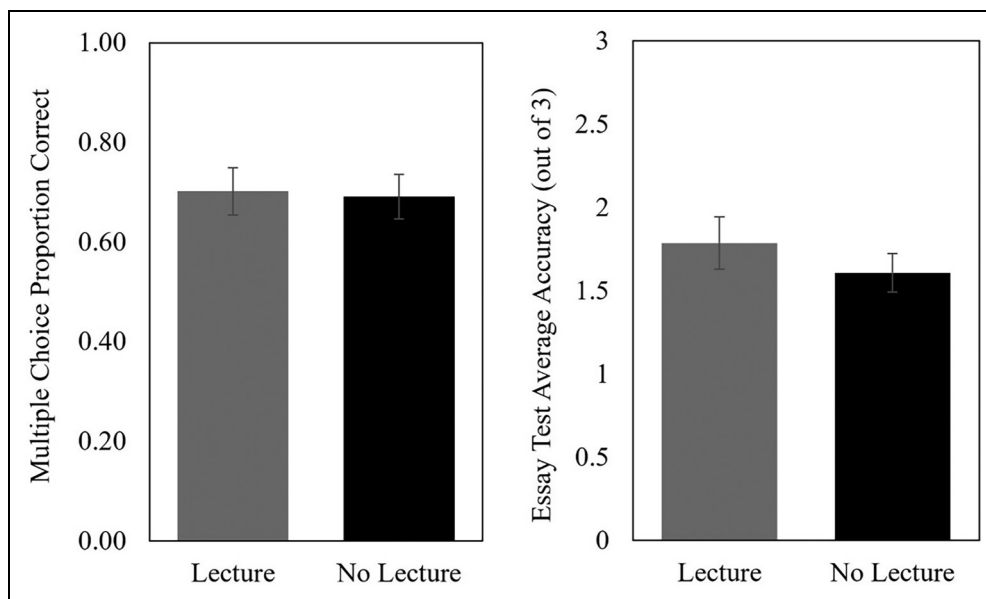


Figure 2. Performance on the final applied multiple choice and essay tests. Note. No significant differences were found between the lecture and no lecture conditions on either the multiple choice or essay tests ($p > .05$). Error bars represent standard errors of the mean.

Discussion

The goal of the present study was to determine whether asynchronous spaced retrieval could effectively deliver foundational course content in the absence of a traditional lecture, thereby providing an innovative option for flipped classrooms. Participants watched a video lecture on a psychology topic and then, on their own time, used a spaced-retrieval app to answer questions about the lecture content and a novel, nonlectured psychology topic for 4 weeks. Afterward, they completed a test of memory transfer consisting of new multiple-choice and short-essay questions. Importantly, spaced retrieval alone appeared to successfully convey course content without a preceding lecture. Participants performed equally well on app questions associated with lectured versus nonlectured content and successfully applied their knowledge of both content areas on the final memory test, demonstrating knowledge transfer rather than mere memorization (cf. Carpenter, 2012). Though participants spent more time reviewing nonlectured versus lectured questions on the app (9.05 vs. 7.92 min per week), they spent less overall time learning the nonlectured content via the app ($M = 36.20$ min) than they would have spent learning the content from the video lecture, had it been delivered per usual in a 50min class.

A potential criticism of the present study is that we did not include a control group in which participants only watched a lecture before taking the final memory test 4 weeks later. This would allow a direct comparison of memory performance in the conventional learning paradigm (lecture only) to learn via spaced retrieval only. We chose not to include that control group for two reasons. First, this group was recently examined in a similar experiment. McHugh et al. (2021) had participants watch a 23min, introductory-level lecture about the country

Georgia. Some learners then used Blank Slate for 4 weeks to study the lecture content, while others did no form of review. At the end of 4 weeks, both groups took a 60-item multiple-choice test about the lecture. Students who watched the lecture but did not review the material scored 57% correct on the final memory test, compared to 95% accuracy for those who used the app to reinforce their learning. Of course, cross-study comparisons should be made with caution, but that number is, expectedly, lower than students' accuracy on our final test (69%) after using the app for 4 weeks. In addition, the lecture-only comparison was not necessary for the purpose of this pilot. Only the two conditions present were necessary for addressing the question of whether a lecture would enhance students' learning of material beyond what the spaced retrieval app would do.

Another possible limitation of the present study was the lack of an attention check when participants watched the lecture video. Though we did not include diagnostic attention checks, we did require participants to take handwritten notes during the lecture, and participants were supervised by a research assistant during this phase. Additionally, as shown in the learning curves in Figure 1, the data support the assumption that participants were paying attention to the lecture—they scored nearly 20 percentage points higher on lectured versus nonlectured questions when they first started using the app. Finally, it should be noted that we did not control whether participants accessed outside sources when answering questions. Future research should poll students about whether they used such sources or engaged in additional studying outside of app use. We believe this was unlikely because only question completion, not accuracy, was incentivized.

By nature of being a pilot study, there are several ways to improve and expand upon this work before any strong recommendations for classroom implementation can be made.

Replication in an authentic classroom setting will be important for establishing ecological validity, as the current study employed paid participants. Relatedly, future research must determine whether these results will replicate with learning material that is more advanced or complex than introductory psychology concepts. In addition, considering our 1-day interval between the end of app use and the final memory test, it remains to be determined whether memories of lectured material would be more durable over time than those acquired via retrieval practice alone. Finally, though the motivation of this study was to reduce time burdens on instructors and students, it must be acknowledged that it does take resources to employ spaced retrieval practice via an app. For instance, there is an up-front time cost for instructors to create questions or select them from banks, and there is a potential financial cost to students depending on the spaced retrieval platform that is used. On that latter point, although this particular app was indeed effective, delivering foundational course content via spaced retrieval practice can be done in any number of ways, including using free electronic resources or even paper-based tasks like flashcards.

In conclusion, our findings offer preliminary insights into potential practical applications for instructors. In the present study, each lecture video we recorded presented content that the first author typically covers in a 50min classroom lecture, accounting for time spent on other activities that occur as part of a live class meeting. Previous research suggests that in the conventional learning paradigm, students investing a 50min class period to get exposed to lecture content might achieve about 57% accuracy on a multiple-choice test administered 4 weeks later (McHugh et al., 2021). Our findings indicate that an alternative learning paradigm involving a spaced retrieval app could allow students to learn the same material asynchronously, investing approximately 36 min spread over time rather than at once, and achieve around 69% accuracy on an applied knowledge test. Thus, this approach saves 28% of time spent learning and improves test performance by 21%, which could take students from failing to a passing grade on a given assessment. It is important to note that these findings do not imply that spaced retrieval should replace all lecture content. Instead, this method may be strategically used to complement traditional teaching, enabling instructors to save in-class time to focus on more advanced concepts and discussions. At the same time, students may improve the efficiency and efficacy of their independent study efforts using evidence-based methods.

Declaration of Conflicting Interests

Amy Smith is an employee at Blank Slate Technologies, which owns the mobile application used during the learning phase of the present study. To avoid any potential conflicts of interest in this research, Amy Smith was not involved in data collection, all analyses were independently verified by first author Leamarie Gordon, and all authors reviewed and approved the final version of this submission.

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Open Practices



For publishing their data, Gordon et al. received badges for open data. The public content may be retrieved from <https://osf.io/54qgs/>

References

- Carpenter, S. K. (2012). Testing enhances the transfer of learning. *Current Directions in Psychological Science*, 21(5), 279–283. <https://doi.org/10.1177/0963721412452728>
- Cohen, J. (1973). Eta-squared and partial eta-squared in fixed factor ANOVA designs. *Educational and Psychological Measurement*, 33(1), 107–112. <https://doi.org/10.1177/001316447303300111>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>
- Gordon, L. T., Hughes, G. I., & Smith, A. S. (2025). *Reducing traditional lecture time through asynchronous spaced retrieval*. Open Science Framework. <https://osf.io/54qgs/>
- Higham, P. A., Zengel, B., Bartlett, L. K., & Hadwin, J. A. (2022). The benefits of successive relearning on multiple learning outcomes. *Journal of Educational Psychology*, 114(5), 928–944. <https://doi.org/10.1037/edu0000693>
- McHugh, D., Feinn, R., McIlvenna, J., & Trevithick, M. (2021). A random controlled trial to examine the efficacy of Blank Slate: A novel spaced retrieval tool with real-time learning analytics. *Education Sciences*, 11(3), 90. <https://doi.org/10.3390/educsci11030090>
- O'Flaherty, J., & Phillips, C. (2015). The use of flipped classrooms in higher education: A scoping review. *The Internet and Higher Education*, 25, 85–95. <https://doi.org/10.1016/j.iheduc.2015.02.002>
- Pan, S. C., & Rickard, T. C. (2018). Transfer of test-enhanced learning: Meta-analytic review and synthesis. *Psychological Bulletin*, 144(7), 710–756. <https://doi.org/10.1037/bul0000151>
- Spielman, R. M., Dumper, K., Jenkins, W., Lacombe, A., Lovett, M., & Perlmutter, M. (2020). *Psychology*. Lumen Learning. Retrieved from <https://courses.lumenlearning.com/suny-hvcc-psychology-1>
- Vaughn, K. E., & Rawson, K. A. (2011). Diagnosing criterion-level effects on memory: What aspects of memory are enhanced by repeated retrieval? *Psychological Science*, 22(9), 1127–1131. <https://doi.org/10.1177/095679761141772>
- Vaughn, K. E., & Rawson, K. A. (2014). Effects of criterion level on associative memory: Evidence for associative asymmetry. *Journal of Memory and Language*, 75, 14–26. <https://doi.org/10.1016/j.jml.2014.04.004>
- Wissman, K. T., Rawson, K. A., & Pyc, M. A. (2012). How and when do students use flashcards? *Memory*, 20(6), 568–579. <https://doi.org/10.1080/09658211.2012.687052>
- Zung, I., Imundo, M. N., & Pan, S. C. (2022). How do college students use digital flashcards during self-regulated learning? *Memory*, 30(8), 923–941. <https://doi.org/10.1080/09658211.2022.2058553>